MinMaxObserver#

https://pytorch.org/docs/stable/generated/torch.ao.quantization.observer.Mir

Description:

Observer module for computing the quantization parameters based on the running min and max values. This observer uses the tensor min/max statistics to compute the quantization parameters. The module records the running minimum and maximum of incoming tensors, and uses this statistic to compute the quantization parameters. dtype – dtype argument to the quantize node needed to implement the reference model spec. qscheme – Quantization scheme to be used reduce_range – Reduces the range of the quantized data type by 1 bit

Function Signature

```
class
torch.ao.quantization.observer.MinMaxObserver(dtype=torch.qu
int8, qscheme=torch.per_tensor_affine, reduce_range=False,
quant_min=None, quant_max=None, factory_kwargs=None,
eps=1.1920928955078125e-07, is_dynamic=False, **kwargs)
[source]#

Parameters

calculate_qparams()[source]#

forward(x_orig)[source]#

reset_min_max_vals()[source]#
```

Notes & Warnings

⚠ WARNING Warning dtype can only take torch.qint8 or torch.quint8.

NOTE Note If the running minimum equals to the running maximum, the scale and zero_point are set to 1.0 and 0.

Detailed Documentation

Documentation

Observer module for computing the quantization parameters based on the running min and max values.

This observer uses the tensor min/max statistics to compute the quantization parameters. The module records the running minimum and maximum of incoming tensors, and uses this statistic to compute the quantization parameters.

dtype – dtype argument to the quantize node needed to implement the reference model spec.

qscheme – Quantization scheme to be used

reduce_range – Reduces the range of the quantized data type by 1 bit

quant_min – Minimum quantization value. If unspecified, it will follow the 8-bit setup.

quant_max – Maximum quantization value. If unspecified, it will follow the 8-bit setup.

eps (Tensor) – Epsilon value for float32, Defaults to torch.finfo(torch.float32).eps.

Given running min/max as xminx_\text{min}xmin and xmaxx_\text{max}xmax, scale sss and zero point zzz are computed as:

The running minimum/maximum xmin/maxx_\text{min/max}xmin/max is computed as:

where XXX is the observed tensor.

The scale sss and zero point zzz are then computed as:

where QminQ_\text{min}Qmin and QmaxQ_\text{max}Qmax are the minimum and maximum of the quantized data type.

dtype can only take torch.qint8 or torch.quint8.

If the running minimum equals to the running maximum, the scale and zero_point are set to 1.0 and 0.

Calculates the quantization parameters.

Records the running minimum and maximum of x.

Resets the min/max values.